Sahaj Bhadja

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Summary

Systems-oriented software engineer with 2+ years building **C++/Python backends, concurrent pipelines, and distributed applications**; experienced in Linux, performance optimization, fault tolerance, simulation, testing, cloud infrastructure, and autonomous-driving research.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Operating Systems, Object-Oriented Programming, Software Engineering, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Engineered **fault-tolerant distributed Python services** processing 1,000+ concurrent jobs with explicit states, failure isolation, and recovery, increasing throughput 3x while preventing cascading failures across production workloads.
- Optimized PostgreSQL and SQL Server access across 10,000+ records using indexing, validation, and performance diagnostics, reducing latency 35% while sustaining 99%+ reliability across 5,000+ recurring monthly workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed a **multithreaded Python processing engine** using ThreadPoolExecutor, secure file ingestion, and fault-tolerant execution, handling 1,000+ files per batch and 10,000+ daily requests with stable peak-load performance.
- Owned backend features through design, implementation, testing, deployment, monitoring, and complex issue debugging, reducing feedback turnaround 60% while supporting reliable production operations for 150+ instructors during peak loads.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **autonomous-driving simulation and data pipelines** over 100,000+ trajectories, translating behavioral-planning requirements into production-quality Python experiments that improved long-horizon planning performance 18% across controlled simulation evaluations.
- Designed a spatial-temporal model combining road geometry, lane direction, and historical trajectories through multi-head attention, improving closed-loop stability 15% while evaluating safety-relevant behavior across 10,000 driving episodes.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Developed **cross-platform Java and Python services** using OOP design patterns, SQL, caching, and API integrations across Linux, Windows, and macOS, reducing response times 30% and technical debt 20%.
- Automated builds and 20+ unit and integration tests with Jenkins, PyTest, Mockito, and Git, generating failure diagnostics while collaborating with Microsoft and GitHub engineers across global Agile sprints.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Built **highly available distributed application services** using Node.js, REST APIs, WebSockets, and Azure SQL, securely synchronizing 2M+ records through eight modular services with role-based access controls across locations.
- Improved system scalability and reliability through Docker packaging, Jenkins delivery pipelines, automated testing, caching, and production diagnostics, achieving 95%+ build validation while reducing deployment time 50% across production releases.

Skills

Languages: C++, Python, Java, JavaScript, TypeScript, SQL, Bash, Shell
Systems: Linux/Unix, operating systems, memory management, networking, TCP/IP, concurrency, multithreading, low-level interactions
Architecture: Distributed systems, fault tolerance, high availability, low-latency systems, microservices, REST APIs, WebSockets
Performance: Profiling, debugging, system diagnostics, performance optimization, caching, indexing, scalability, reliability
Delivery: Docker, Kubernetes, AWS, Azure, Git, Jenkins, GitHub Actions, CI/CD, automated testing, code reviews, Agile

Projects

CPU Scheduling Simulator Live | *C++, React.js, Operating Systems*

- Implemented and benchmarked **six C++ scheduling algorithms** with interactive controls, Gantt visualization, comprehensive edge-case handling, and dynamic metrics across 1,000+ process combinations, reducing average turnaround time 25%.

Chat2SQL Code | *Java, TypeScript, Docker, Linux*

- Enhanced a **cross-platform systems application** across Java and TypeScript components, supporting 10+ databases with structured execution, validation, Docker, Maven, Node.js, and Linux-compatible development and debugging workflows across operating environments.

FinNexus Live | *Kubernetes, MongoDB, WebSockets*

- Built a **distributed real-time platform** supporting 50,000+ users and 1M+ transactions with secure integrations, WebSocket updates, MongoDB persistence, advanced filtering, and Kubernetes-based scalable deployment automation for production workloads.